

Research Gap Analysis

Systematic Performance Evaluation of Standardized Post-Quantum Cryptography on the RP2350 / Hazard3 Architecture

Version: Working Research Notes

Purpose: Consolidated record of all research gaps identified during the literature review. This document distinguishes between validated research gaps, supporting gaps, and rejected gaps to ensure that the final manuscript makes technically accurate and defensible novelty claims.

1. Overview

As the literature review evolved, several initial assumptions were revised after reviewing recent work (including pqm4, Liu et al., Schöffel et al., Fitzgibbon & Ottaviani, Nigam et al., Chhetri, and recent RISC-V studies).

Rather than relying on a single novelty claim, the paper should be motivated by a combination of complementary research gaps spanning embedded architectures, benchmarking methodology, comparative evaluation, and practical deployment.

2. Rejected Research Gaps (Do Not Use)

These gaps were initially considered but are no longer valid based on the reviewed literature.

RG-R1 --- No benchmarking exists on the Raspberry Pi Pico 2 W / RP2350

Status: Rejected

Nigam et al. benchmarked ML-KEM on the RP2350A using the Arm Cortex-M33 architecture.

RG-R2 --- No benchmarking exists on ARM Cortex-M33

Status: Rejected

Multiple recent studies evaluate PQC on Cortex-M33 platforms.

RG-R3 --- No benchmarking exists for ML-KEM

Status: Rejected

ML-KEM is among the most extensively evaluated standardized PQC

algorithms.

RG-R4 --- No benchmarking exists for standardized PQC algorithms

Status: Rejected

Recent work evaluates standardized ML-KEM and ML-DSA implementations.

3. Valid Supporting Research Gaps

RG-1 --- Fragmented Algorithm Coverage

Priority: High

Most studies evaluate only one algorithm or a limited subset of standardized algorithms (e.g., ML-KEM only, or ML-KEM + ML-DSA). Few provide a broad comparative evaluation under identical conditions.

Why it matters - Prevents consistent comparison. - Makes algorithm selection difficult. - Limits deployment guidance.

RG-2 --- Lack of a Unified Benchmarking

Methodology

Priority: Very High

Existing studies differ in: - Hardware platforms - Compiler options - Optimization levels - Benchmarking frameworks - Implementation libraries - Evaluation metrics

Consequently, benchmarking results from different studies cannot be directly compared.

Importance This is one of the strongest and most defensible research gaps.

RG-3 --- Platform Diversity Without Standardized Evaluation

Priority: High

Benchmarking has been conducted on: - ARM Cortex-M0+ - ARM Cortex-M4 - ARM Cortex-M33 - Raspberry Pi 4 - Generic RISC-V - FPGA-based accelerators - Custom RISC-V ISA extensions

However, the diversity of hardware platforms makes cross-platform comparisons difficult.

RG-4 --- Limited Deployment-Oriented Analysis

Priority: Very High

Most studies report: - Execution time - Memory usage - Energy

consumption

Few translate these measurements into practical recommendations for embedded system designers.

The literature rarely answers:

Which standardized PQC algorithm is the most suitable for deployment on a constrained IoT platform?

RG-5 --- Limited Systematic Comparison of Standardized NIST Algorithms

Priority: Very High

Although standardized algorithms are increasingly benchmarked, comprehensive comparisons conducted under identical experimental conditions remain limited.

This differs from RG-1 because the focus here is consistency of evaluation rather than simply algorithm coverage.

4. Architecture-Centric Research Gaps

These gaps become the foundation if the paper pivots toward evaluating the Hazard3 RISC-V architecture.

RG-6 --- Limited Evaluation of the Hazard3

RISC-V Architecture

Priority: Very High

Existing embedded benchmarking literature is heavily ARM-centric.

Published RISC-V studies primarily investigate: - FPGA SoCs - Hardware accelerators - ISA extensions - Research processors

Limited published work evaluates standardized PQC algorithms on the lightweight Hazard3 RISC-V cores integrated into a commercially available microcontroller.

Note: This claim should continue to be verified before publication.

RG-7 --- Lack of Controlled ARM vs. Hazard3 Comparison

Priority: Extremely High

The RP2350 uniquely integrates: - Dual Arm Cortex-M33 - Dual Hazard3 RISC-V

on the same silicon.

This enables an architecture comparison while holding constant: - SRAM - Flash - Clock frequency - Peripheral subsystem - Manufacturing process

Such a controlled experimental design is rare in the existing literature.

RG-8 --- Limited Architectural Analysis of PQC Workloads

Priority: High

Most studies report performance numbers but do not analyze architectural behaviour.

Different PQC families stress processors differently:

Lattice-based - Number Theoretic Transform (NTT) - Modular arithmetic
- Polynomial multiplication

Hash-based - SHAKE / Keccak - Merkle tree operations

Understanding these workload characteristics can provide deeper architectural insights.

5. Experimental Methodology Gaps

RG-9 --- Need for Reproducible Benchmarking

Priority: Medium--High

Existing studies use: - pqm4 - PQCclean - liboqs - Rust implementations - Assembly optimizations - Custom software

A reproducible benchmarking methodology improves fairness and repeatability.

RG-10 --- Comprehensive Metric Collection

Priority: High

Many studies evaluate only a subset of: - Execution time - CPU cycles - RAM - Flash - Stack - Binary size - Energy

Collecting these metrics under a unified methodology provides a more complete assessment.

6. Recommended Core Research Gaps

These should form the foundation of the paper.

1. **RG-6** --- Limited evaluation of standardized PQC on the Hazard3 RISC-V architecture.
 2. **RG-7** --- Lack of controlled ARM Cortex-M33 vs. Hazard3 comparison on identical hardware.
 3. **RG-2** --- Lack of a unified benchmarking methodology.
 4. **RG-5** --- Limited systematic comparison of multiple standardized NIST PQC algorithms.
 5. **RG-4** --- Limited deployment-oriented guidance for embedded IoT systems.
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7. Proposed Research Direction

Rather than positioning the work as:

Benchmarking PQC on the Raspberry Pi Pico 2 W

the stronger positioning is:

Evaluating the suitability of the Hazard3 RISC-V architecture for deploying standardized NIST post-quantum cryptographic algorithms in resource-constrained IoT systems through a systematic and reproducible benchmarking methodology.

In this framing: - **The architecture** is the research subject. - **The standardized PQC algorithms** are the workloads. - **The RP2350** is the experimental platform.

This positioning aligns the work with embedded systems, computer architecture, IoT security, and applied post-quantum cryptography.